



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information

Course Code: **CSC103**

Credit Hours: **4 (3, 1)**

Lab Hours/Week: **3**

Course Title: **Programming Fundamentals**

Lecture Hours/Week: **3**

Pre-Requisites: **None**

Catalogue Description:

This course emphasizes the basic concepts used in programming. The topics include: Computer Programming; Basic Syntax & Semantics of a Higher-Level Language; Conditional & Iterative Control Structures; Functions & Parameter Passing; Recursion; Arrays; String Processing; Exception Handling; and File I/O.

Unit wise Major Topics:

Unit	Topic	No. of teaching hours
1.	Computer Programming: Fundamental Concepts and Programming Paradigm.	3
2.	Basic Syntax & Semantics of a Higher-Level Language: Program Style & Documentation; Variables & Primitive Data Types; Expressions & Assignments; and Simple I/O.	7.5
3.	Conditional & Iterative Control Structures.	9
4.	Functions & Parameter Passing; and Recursion.	9
5.	Arrays; String Processing,	9
6.	Exception Handling, and File I/O	7.5
Total Contact Hours		45

Mapping of CLOs and SOs

Sr.#	Unit #	Course Learning Outcomes	Blooms Taxonomy Learning Level	SO
CLO's for Theory				
CLO-1	1-2	Demonstrate the fundamental concepts of programming and	<i>Understanding</i>	1
CLO-2	3-5	Employ programming constructs using a programming language	<i>Applying</i>	2,4
CLO-3	6	Handle programs utilizing exception and file I/O.	<i>Applying</i>	2,4
CLO's for Lab				
CLO-4	2-6	Implement a program using programming constructs.	<i>Applying</i>	2,4
CLO-5	1-6	Build a medium size application in a team environment	<i>Creating</i>	2-5

CLO Assessment Mechanism					
Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5
Quizzes	Quiz 1	Quiz 2 & 3	Quiz 4	-	-
Assignments	Assignment 1	Assignment 2&3	Assignment 4	Lab Assignment	-
Mid Term Exam	Mid Term Exam	Mid Term Exam	-	-	-
Final Term Exam	Final Term Exam				
Project	-	-	-	-	Lab Project
Text and Reference Books					
Textbook: 1. Java How to Program, Deitel, P. & Deitel, H., Prentice Hall, 2019.					
Reference Books: 1. Java: The Complete Reference, Herbert Schildt, Prentice Hall, 2018. 2. Introduction to Java Programming and Data Structures, Comprehensive Version, Y.D.Liang, Pearson, 2017. 3. Java: Programming Basics for Absolute Beginners, Nathan Clark, CreateSpace Independent Publishing Platform, 2017.					